

Ajaykrishnan Selucca Muralidharan

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SUMMARY

PhD researcher in **Operations & Maintenance Engineering** at Luleå University of Technology, working on statistical and machine-learning frameworks for **railway wayside detector data quality assessment**. Research is funded by **Europe's Rail JU (TRANS4M-R)** and **Trafikverket/JVTC**, conducted from the infrastructure manager's perspective. Core focus: context-aware quantile analysis, seasonal-trend decomposition, and ML-based residualisation of WILD, HBD, and HWD detector datasets on the Malmbanan heavy-haul corridor. Prior industry experience in production ML systems (recommendation engines, cloud deployment) at scale.

FELLOWSHIPS & AWARDS

2026 **JSPS Summer Program Fellowship** - Japan Society for the Promotion of Science
Fellowship ID: SP26501 | Host: Prof. Masafumi Miyatake, Sophia University, Tokyo
June 15 - August 25, 2026. Competitive international fellowship for doctoral researchers; selected based on research proposal submitted.

WORK EXPERIENCE

PhD Researcher - Luleå University of Technology (LTU) May 2024 - Present

- **Funding:** EU-Rail TRANS4M-R (multi-national initiative, 2022-2026) and Trafikverket/JVTC (Swedish Transport Administration), focused on standardised European railway condition monitoring.
- **Research Framework:** Four-layer statistical and ML framework for wayside detector data quality: (1) quantile characterisation, (2) seasonal-trend decomposition, (3) ML residualisation, (4) ontology-based integration.
- **WILD Analytics:** Context-aware quantile scoring (q99.5 with bootstrap CIs), bimodality analysis, cross-site normalisation, and drift/anomaly detection across the Malmbanan corridor.
- **HBD/HWD Modelling:** Seasonal-contextual thermal baseline modelling; identified $\sim 10^{\circ}\text{C}$ bearing independent seasonal amplitude and $3.6\times$ over-flagging bias under fixed thresholds.
- **Fiber-Optic Integration:** DAS signal alignment with WILD events for cross-validation and calibration; building scalable processing workflows.
- **Tools:** Python (NumPy, Pandas, Statsmodels, Scikit-learn, XGBoost), Git, Docker, LaTeX.

Junior Associate - Primesoft Enterprise IT Services Aug 2022 - May 2024

- Delivered ML-driven personalisation and churn solutions for OTT clients as modular Flask APIs.
- Built a production *“Because You Watched”* hybrid recommender (content-based + collaborative) integrated with Couchbase/Datastore; containerised and deployed on GCP (Cloud Run, GKE).
- Implemented churn and dashboard-ranking PoCs that clustered users to optimise engagement and row/item ordering.
- **Tech:** Python, Flask, Docker, GCP, Snowflake, BigQuery.

Intern - Applied Materials India Jul 2021 - Dec 2021

- Developed stabilisation-detection models using sensor-based time-series learning in semiconductor manufacturing tools.

EDUCATION

2024 - Present	PhD, Operations & Maintenance Engineering , Luleå University of Technology (Sweden) <i>Licentiate thesis in preparation (expected: November 2026).</i> Supervisors: Prof. Matti Rantatalo, Johan Odelius, Florian Thiery, Praneeth Chandran
2020 - 2022	M.Tech., Artificial Intelligence , Amrita Vishwa Vidyapeetham (India) <i>Thesis: Real-Time Health Monitoring of Bolted Joints in Machines using Predictive Analytics.</i>
2019 - 2021	Post Graduate Program in Artificial Intelligence and Machine Learning <i>Great Lakes Institute of Management (India) & McCombs School of Business, UT Austin (USA).</i>
2018 - 2019	B.E., Electronics & Communication Engineering , St. Peter's Institute of Higher Education & Research (India)

PUBLICATIONS

Conference Papers

- Selucca Muralidharan, A., Thiery, F., Chandran, P., Odelius, J., & Rantatalo, M. (2026). *Statistical analysis of hot-box and hot-wheel detector data for context-aware rolling-stock monitoring*. Presented at *Transport Research Arena (TRA) 2026*, Budapest.
- Thiery, F., Selucca Muralidharan, A., Chandran, P., Odelius, J., & Rantatalo, M. (2026). *Standardizing wayside monitoring data via CDF rescaling: A case study on Wheel Impact Load Detectors*. Presented at *Transport Research Arena (TRA) 2026*, Budapest.
- Muralidharan, A.S., Thiery, F., Chandran, P., Odelius, J., & Rantatalo, M. (2026). *Investigating the quality of Wheel Impact Load Detector data for building predictive maintenance strategies: A visualization and statistical approach*. In: Kour, R., Karim, R., Kumar, U., Galar, D., Jägare, V. (eds) *IAI 2025. Lecture Notes in Mechanical Engineering*. Springer, Cham. https://doi.org/10.1007/978-3-032-03725-1_25

Theses

- Selucca Muralidharan, A. (2022). *Real-time health monitoring of bolted joints in machines using predictive analytics* (M.Tech. thesis, Amrita Vishwa Vidyapeetham, India).

SKILLS

Statistical Methods	Time-series decomposition (STL, STR), quantile regression, bootstrap inference, distribution fitting, non-parametric tests, uncertainty analysis
Machine Learning	Anomaly detection, clustering, regression, feature engineering, model evaluation
Programming	Python (NumPy, Pandas, Scikit-learn, Statsmodels, PyTorch), SQL, PySpark
Data & Signal	Sensor data processing, context-aware modelling, railway detector analytics, visualisation (Matplotlib, Seaborn, Plotly)
Research Practices	LaTeX, reproducible notebooks, Git, Docker, structured academic writing
Computing	GCP (BigQuery, Cloud Run, GKE), Snowflake, HPC/Linux environments

CERTIFICATIONS

- Nov 2024 **Neo4j Certified Professional** - [Credential Link](#)
Apr 2022 **Microsoft Certified: Azure AI Fundamentals (AI-900)** - [Credential Link](#)

REFERENCES

Prof. Matti Rantatalo

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Dr. Florian Thiery

Senior Lecturer (co-supervisor), Division of Operation and Maintenance Engineering, LTU
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